

Under review as a conference paper at ICAIS 2025

000
001
002
003
004
005
006
007
008
009
010
011
012
013
014
015
016
017
018
019
020
021
022
023
024
025
026
027
028
029
030
031
032
033
034
035
036
037
038
039
040
041
042
043
044
045
046
047
048
049
050
051
052

ENHANCING AI CONFERENCE PEER REVIEW QUALITY THROUGH ANONYMIZED FEEDBACK AND ADAPTIVE REWARD SYSTEMS

Anonymous authors
Paper under double-blind review

ABSTRACT

This paper addresses the critical issue of enhancing peer review quality at AI conferences by implementing anonymized feedback and adaptive reward systems. The growing volume of conference submissions and limited reviewer accountability result in inconsistent review quality, bias, and a lack of transparency, posing significant challenges to the integrity of AI research. Our proposed solution involves a dynamic feedback loop that anonymizes and aggregates feedback to minimize biases, coupled with an adaptive reward system to motivate reviewers while preserving the integrity of the review process. Utilizing sentiment analysis, feedback is processed to detect and mitigate potential biases, enhancing the fairness and efficacy of peer reviews. Experiments conducted using a logistic regression model on the Yelp Polarity dataset demonstrate a significant improvement in sentiment classification accuracy, from 54.1% to 83.4%, indicating the effectiveness of our anonymized feedback loop. However, the bias detection score of 0.0 across all runs highlights the need for further refinement in bias mitigation. Our method's scalability and adaptability across various conference settings are supported by its successful implementation in sentiment analysis tasks. Overall, this study provides a robust framework for enhancing the accountability and quality of peer reviews, with implications for future research aimed at integrating advanced bias detection and mitigation techniques.

1 INTRODUCTION

The peer review process serves as a cornerstone of scientific integrity and quality assurance within the AI research community. However, the exponential growth in AI conference submissions, combined with limited reviewer accountability, has led to significant challenges such as inconsistent review quality, bias, and a lack of transparency (Samuels & Mcgonical, 2020; Denis et al., 2013; Bergstrom & Gross, 2025). These issues threaten the scientific advancement of AI research, posing a critical question: *Can innovative methodologies like anonymized feedback and adaptive reward systems enhance the accountability and quality of peer reviews at AI conferences?* Studies have explored the role of generative AI tools in enhancing the peer review process by providing effective feedback (Lim et al., 2025; Berrezueta-Guzman et al., 2025).

Addressing these inefficiencies and biases is paramount for maintaining the integrity of scientific research and fostering innovation. The demand from the research community for solutions that can scale with the growing complexity of AI conferences is urgent, as existing systems are increasingly inadequate (Fazil et al., 2024; Feldman & Peake, 2021; Kim et al., 2025). Recent developments in sentiment analysis and bias detection offer promising methodologies that can be integrated into peer review systems (Dervisoglu & Amasyali, 2021; R. et al., 2025; Duraisamy et al., 2022; Beasley et al., 2020). Enhancing the peer review process could lead to fairer and more rigorous assessments, ultimately improving research credibility and accelerating scientific discovery (Ahad, 2023).

Despite the clear need for reform, improving the peer review process presents inherent challenges. Naive approaches often fail to address the nuanced dynamics of peer reviews, where biases may stem



Meta Review Status: The meta review and decision have not been released yet.

AI Review from DeepReviewer



AI Review from DeepReviewer will be automatically processed



Summary

This paper introduces a framework aimed at enhancing the peer review process for AI conferences, addressing critical issues such as inconsistent review quality, bias, and lack of transparency. The authors propose a system that integrates anonymized feedback and adaptive reward mechanisms to improve the fairness and effectiveness of reviews. The core idea revolves around a dynamic feedback loop where reviewer comments are anonymized and aggregated, minimizing potential biases. This is coupled with a tiered reward system designed to incentivize high-quality reviews, where rewards are allocated based on the quality of the review and the specific context of the conference. To achieve this, the authors employ machine learning techniques, specifically sentiment analysis, to process and optimize the feedback. The framework's effectiveness was evaluated using a logistic regression model trained on the Yelp Polarity dataset, a benchmark dataset in sentiment analysis. The experiments demonstrated a significant improvement in sentiment classification accuracy, suggesting that the anonymized feedback loop can effectively enhance the quality of review processing. However, the authors acknowledge that the bias detection component of their framework requires further refinement, as it did not perform well in the initial experiments. The paper concludes by discussing the potential implications of their framework for future peer review processes, highlighting its adaptability and scalability for use in various conferences and disciplines. The authors emphasize that their approach could lead to more objective and transparent review processes, ultimately contributing to the integrity of scientific research and fostering innovation within the AI community. The framework's potential to be implemented across various conferences and disciplines is a significant contribution to the broader academic community, offering a structured approach to improve the often subjective and inconsistent peer review process. Despite the promising results, the authors acknowledge the need for further research to address the limitations of their approach, particularly in the area of bias detection and mitigation.



Strengths

The paper's primary strength lies in its focus on a critical and timely issue within the academic community: the need to improve the peer review process, particularly in the context of AI conferences. The authors have identified a genuine problem, where the increasing volume of submissions and limited reviewer accountability can lead to inconsistent review quality and bias. By proposing a framework that integrates anonymized feedback and adaptive reward systems, they offer a

potential solution to enhance the fairness and effectiveness of the peer review process. The use of machine learning techniques, such as sentiment analysis, to process and optimize feedback is an innovative approach that could lead to more objective and transparent review processes. The framework's scalability and adaptability suggest that it could be implemented across various conferences and disciplines, making it a valuable contribution to the broader academic community. The authors' emphasis on improving the accountability and quality of peer reviews has implications for maintaining the integrity of scientific research and fostering innovation in the AI community. The experimental results, while limited to a sentiment analysis task, demonstrate the potential of the anonymized feedback loop to improve the quality of review processing. The authors have clearly articulated the problem they are trying to solve and have proposed a novel framework that addresses this problem. The paper's focus on enhancing the peer review process is a significant contribution to the academic community, and the proposed framework has the potential to improve the fairness and effectiveness of peer reviews.

Weaknesses

After a thorough examination of the paper, I've identified several key weaknesses that significantly impact the validity and generalizability of the proposed framework. First, the experimental validation is limited by the use of the Yelp Polarity dataset, which consists of product reviews and does not accurately represent the complexity and nuances of peer reviews in AI conferences. As the paper states, "We utilize the Yelp Polarity dataset, a benchmark in sentiment analysis, to emulate the sentiment analysis component of our feedback processing system." This choice is problematic because the language, technical depth, and critical evaluation present in academic peer reviews are substantially different from those found in product reviews. This discrepancy raises serious concerns about the generalizability of the sentiment analysis results to the actual target domain of AI conference reviews. The paper does not provide any justification for using this dataset or discuss its limitations in the context of peer review simulation. This is a critical oversight that undermines the validity of the experimental results. My confidence in this weakness is high, as the paper explicitly states the use of the Yelp Polarity dataset and lacks any discussion of its suitability for the task. Second, the paper lacks real-world experiments or case studies in a conference context, which severely limits the generalizability of the findings. The entire "EXPERIMENTS" section focuses on the sentiment analysis task using the Yelp Polarity dataset. There is no mention of experiments conducted with actual peer review data or in a simulated conference setting. This absence of real-world validation makes it difficult to assess the practical effectiveness and generalizability of the proposed framework in the intended context of AI conference peer reviews. The paper does not cite any studies that have implemented similar systems in real-world conference settings, further highlighting the lack of empirical support for the proposed framework. This limitation significantly reduces my confidence in the practical applicability of the proposed system. My confidence in this weakness is high, as the paper's experimental section clearly focuses on the Yelp Polarity dataset and lacks any real-world validation. Third, the paper does not provide sufficient detail on how the proposed framework would handle various practical challenges, such as reviewer recruitment, training, and retention. The "METHODOLOGY" section describes the anonymized feedback mechanism and adaptive reward system but does not detail processes for reviewer recruitment, training, or retention. The "Adaptive Reward System" subsection focuses on the reward calculation based on review quality and conference context but doesn't address how reviewers are initially recruited or kept engaged. This lack of detail raises concerns about the practical feasibility and adoption of the proposed framework in real-world conference settings. The paper focuses on the technical aspects of feedback anonymization and reward allocation but neglects the practical implementation challenges related to managing reviewers. My confidence in this weakness is high, as the paper's methodology section lacks any discussion of these practical challenges. Fourth, the evaluation metrics used in the experiments may not capture all aspects of review quality, and the paper does not discuss alternative metrics or methods for evaluating the effectiveness of the proposed system. The

"Evaluation Metrics" subsection states: "Accuracy is the primary metric, representing the percentage of correct predictions. We also introduce a bias detection score as a secondary metric to assess the fairness of sentiment processing." The paper does not discuss other potential metrics for evaluating review quality, such as clarity, thoroughness, or constructiveness. Relying solely on sentiment analysis accuracy might not fully capture the multifaceted nature of review quality. The lack of discussion on alternative metrics limits the assessment of the framework's overall effectiveness in improving peer reviews. My confidence in this weakness is high, as the paper explicitly states the use of only accuracy and bias detection score as evaluation metrics. Fifth, the paper does not address the potential for gaming or manipulation of the reward system by reviewers, which could undermine the integrity of the review process. The "Adaptive Reward System" subsection describes the reward calculation based on review quality and conference context but does not include mechanisms to prevent reviewers from strategically manipulating their reviews to maximize rewards. There is no mention of checks for consistency, validity, or potential gaming strategies. The absence of mechanisms to prevent gaming raises concerns about the robustness and integrity of the proposed reward system. My confidence in this weakness is high, as the paper's description of the reward system lacks any discussion of preventing manipulation. Sixth, the absence of a discussion on the computational resources required to implement the framework limits the practicality of the proposed solution. The "Implementation Details" subsection mentions: "Experiments are conducted in Python with PyTorch, managing dependencies using standard packages." and "Training is performed on a GPU, defaulting to CPU when necessary." However, there is no detailed analysis of the computational resources required for a full-scale implementation, such as server specifications, memory requirements, or processing time for a large number of reviews. The lack of information on computational resources makes it difficult to assess the practical feasibility and scalability of the proposed framework, especially for large conferences. My confidence in this weakness is high, as the paper provides only basic implementation details and lacks a comprehensive discussion of computational resources. Finally, while the paper acknowledges the need for more robust bias detection and mitigation techniques, it does not provide a clear plan for future research to address this and other limitations. The "CONCLUSION" section mentions: "However, a notable limitation is the lack of explicit bias detection, which future work should aim to integrate using advanced techniques (Dervisoglu & Amasyali, 2021; R. et al., 2025; Oyebode, 2025)." While it identifies bias detection as a limitation and suggests future work, it doesn't provide a detailed plan or specific research directions beyond mentioning the need for "advanced techniques." The lack of a clear research plan makes it less clear how the identified limitations will be systematically addressed in future work. My confidence in this weakness is medium, as the paper does acknowledge the limitation but lacks a concrete plan for future research.

Suggestions

To significantly strengthen this paper, I recommend several concrete and actionable improvements. First, the authors should conduct experiments using a dataset that more closely resembles actual peer reviews in AI conferences. This could involve collecting a dataset of anonymized reviews from past conferences, with appropriate permissions and ethical considerations. The dataset should include a variety of review types, such as full reviews, rebuttals, and reviewer comments, to capture the complexity of the peer review process. This would provide a more realistic evaluation of the framework's performance and address the current limitation of using a product review dataset. The authors should also consider using more sophisticated evaluation metrics that capture aspects of review quality beyond sentiment, such as the clarity, thoroughness, and constructiveness of the feedback. For example, metrics could be developed to assess the presence of specific feedback elements, such as suggestions for improvement, identification of limitations, and justification of scores. These metrics could be based on natural language processing techniques, such as topic modeling or semantic similarity analysis. This would provide a more comprehensive assessment of the framework's effectiveness in improving the quality of peer reviews. Second, the authors should provide a detailed discussion of how the proposed framework would handle

practical challenges, such as reviewer recruitment, training, and retention. This could involve developing strategies for incentivizing reviewers to participate in the system, providing training materials to ensure that reviewers understand how to use the system effectively, and implementing mechanisms for retaining reviewers over time. The authors should also address the potential for gaming or manipulation of the reward system by reviewers. This could involve developing mechanisms for detecting and preventing strategic behavior, such as reviewers giving overly positive or negative feedback to manipulate their rewards. For example, the system could incorporate checks for consistency and validity of reviews, and implement penalties for reviewers who engage in malicious behavior. This would enhance the integrity and robustness of the proposed framework. Third, the authors should provide a detailed analysis of the computational resources required to implement the framework, including the hardware and software requirements, and the time complexity of the algorithms. This analysis should consider the scalability of the framework to handle large volumes of submissions and reviews. This would provide a more realistic assessment of the practical feasibility of the proposed solution. Fourth, the authors should provide a clear plan for future research to address the identified limitations, such as the need for more robust bias detection and mitigation techniques. This could involve exploring different machine learning models for bias detection, and developing methods for mitigating bias in the review process. The authors should also consider investigating the impact of the proposed framework on the diversity of accepted papers and the overall quality of the conference. This would provide a more comprehensive understanding of the potential benefits and drawbacks of the proposed system. Finally, the authors should consider piloting their framework in a real-world conference setting to validate its effectiveness and identify any practical challenges that may arise. This would provide valuable insights into the practical implementation of the framework and help to refine it for broader adoption. These suggestions are aimed at addressing the identified weaknesses and enhancing the overall quality and impact of the paper.

? Questions

After reviewing the paper, I have several questions that I believe are crucial for further understanding and development of the proposed framework. First, how do you ensure the privacy and confidentiality of the feedback provided by reviewers in your system? What measures are in place to protect the identities of reviewers and prevent unauthorized access to their feedback? This is a critical concern, as the system relies on anonymized feedback, and any breach of confidentiality could undermine the entire process. Second, how do you define and measure the quality of reviews in your framework? What criteria are used to determine whether a review is good or bad, and how are these criteria incorporated into the reward system? The paper mentions using sentiment analysis, but it's unclear how this translates into a comprehensive assessment of review quality. Third, how do you handle cases where reviewers disagree on the quality of a submission? Does your system have a mechanism for resolving conflicts or discrepancies between reviewers' opinions? This is a common challenge in peer review, and it's important to understand how the proposed framework addresses this issue. Fourth, what are the computational requirements for implementing your framework? How scalable is your approach to handle large volumes of submissions and reviews in real-world conference settings? The paper lacks a detailed analysis of the computational resources needed, which is crucial for assessing the practical feasibility of the system. Fifth, how do you plan to validate the effectiveness of your framework in real-world settings? Do you have plans to collaborate with any conferences or organizations to pilot your system? This is essential for demonstrating the practical applicability of the proposed framework. Sixth, how do you address the potential for bias in the machine learning models used for sentiment analysis and reward optimization? What steps are taken to ensure that these models are fair and unbiased? The paper acknowledges the need for bias mitigation, but it's unclear how this will be achieved in practice. Seventh, what measures are in place to prevent reviewers from gaming the reward system? For example, could reviewers intentionally give overly positive or negative feedback to manipulate their rewards? This is a critical concern, as the reward system could be exploited if not

properly designed. Eighth, how do you plan to incorporate feedback from reviewers and other stakeholders to improve your framework over time? Do you have a mechanism for collecting and incorporating user feedback into the design and implementation of your system? This is important for ensuring that the system remains relevant and effective. Ninth, how does your framework handle cases where reviewers may have conflicts of interest with the authors of the submissions they are reviewing? What mechanisms are in place to identify and manage such conflicts? This is a crucial aspect of ensuring the integrity of the peer review process. Finally, how do you plan to fund the development and implementation of your framework? Are there any plans to seek external funding or partnerships to support your work? This is important for the long-term sustainability of the project. These questions are aimed at clarifying key uncertainties and addressing potential limitations of the proposed framework.

 Scores

Soundness: 1.25

Presentation: 1.25

Contribution: 1.25

Rating: 2.0

AI Review from ZGCA

 AI Review from ZGCA will be automatically processed

 Summary

The paper proposes a framework to improve AI conference peer review quality by (1) anonymizing and aggregating feedback and applying TF-IDF-based sentiment analysis (Eq. 1) to reduce recognizability/retaliatory bias, and (2) introducing an adaptive reward system where rewards depend on review quality and conference context ($R_i = \alpha \cdot Q_i + \beta \cdot C_i$ in Eq. 2), with (3) logistic regression (Eq. 3) used to optimize reward allocation. Experiments use the Yelp Polarity dataset to demonstrate sentiment classification accuracy improvements (from 54.1% to 83.4%), while a bias detection score remains 0.0 across all runs. The paper argues the approach is scalable and adaptable across conferences.

✓ Strengths

- Addresses an important and under-explored problem space: accountability, bias mitigation, and incentives in peer review (Sections 1 and 3).
- Clear modular decomposition of the approach (anonymization + sentiment analysis; adaptive rewards; ML optimization), with simple, interpretable formulas (Eqs. 1–3).
- Provides basic implementation details for the sentiment classifier (TF-IDF features, logistic regression with PyTorch), including optimizer and training setup (Section 4).
- Broad literature awareness and motivation tying peer-review challenges to AI tools and incentive mechanisms (Sections 1–2).

✗ Weaknesses

- Empirical validation is misaligned with the core claims: all experiments are on Yelp Polarity sentiment classification (Section 4), not on peer-review text or reviewer behavior/quality metrics. The paper does not evaluate the framework in the target domain.
- The 'bias detection score' reported as 0.0 across all runs (Section 5) is undefined, lacks methodological description, and provides no diagnostic insight; the paper therefore does not substantiate its bias-mitigation claims.
- Key constructs in the reward system are unspecified: Q_i (review quality) and C_i (conference context) are not operationalized or measured; no empirical evaluation of the reward mechanism (no simulations, no observational/historical analysis).
- Causal linkage between anonymization/sentiment analysis and improved peer-review fairness is asserted but not tested (no ablation of anonymization; no recognizability or retaliation bias measurements; no author-reviewer interaction analysis).
- Reproducibility gaps: no random seeds or hardware details; the referenced table in Section 5 is missing; no ablations or qualitative analysis; unclear what changed across runs to move from 54.1% to 83.4%.
- Choice of Yelp Polarity and small data split (5k/1k/1k) limits external validity; logistic regression with TF-IDF on Yelp typically attains higher accuracy under standard splits, so reported numbers are difficult to interpret.
- Writing is verbose with many tangential citations; several crucial definitions (e.g., bias detection metric, reward-tier labeling from historical data) are missing, which reduces clarity (Sections 3–5).

? Questions

- How is the 'bias detection score' defined and computed? What protected attributes or proxies (e.g., affiliation, seniority, geography) are used, and what is the statistical test or metric (e.g., equalized odds, demographic parity, subgroup AUC)?
- What precise steps constitute the 'anonymized feedback loop' beyond removing identifiers? How does anonymization interact with sentiment processing, and how is recognizability or retaliatory bias operationalized and measured?
- How are Q_i (review quality) and C_i (conference context) concretely defined? For Q_i , do you use AC/meta-review ratings, rubric compliance, depth/constructiveness scores, timeliness, or author feedback? For C_i , which conference variables are included and how are α , β set or learned?
- What features x_i are used in Eq. (3) for reward-tier prediction beyond sentiment? Are there ground-truth labels for reward tiers from historical peer-review data? If not, how are labels derived?
- Why is Yelp Polarity an appropriate proxy for peer-review feedback? Did you attempt experiments on peer-review corpora (e.g., OpenReview comments, ACL rolling review), or at least on scientific text domains?
- What changed across Run 1–3 to yield the reported accuracy increases from 54.1% to 83.4% (Section 5)? Provide seeds, hyperparameters, and the exact modifications per run. Please include the missing table.
- Could you provide ablations isolating the effect of anonymization (on/off), aggregation strategies, and sentiment model choices (e.g., TF-IDF LR vs. fine-tuned transformers) on both accuracy and bias metrics?
- How will you prevent Goodhart's law effects and gaming of the reward system (e.g., reviewers optimizing for sentiment or length rather than substance)? Any safeguards or auditing mechanisms?
- What privacy-preserving measures (e.g., de-identification, secure storage) does the anonymization pipeline implement when handling sensitive peer-review text?

! Limitations

- Domain mismatch: sentiment classification on Yelp reviews does not validate effects on peer-review quality, fairness, or bias mitigation.
- Undefined and uninformative bias metric (bias detection score = 0.0) offers no evidence for bias mitigation in the target domain.
- No empirical evaluation of the adaptive reward system (no simulation, historical analysis, or field trial), and key variables (Q_i , C_i) are not operationalized.
- Potential for gaming and perverse incentives if rewards correlate with superficial proxies (e.g., positive sentiment) rather than review substance.
- Risk of penalizing writing styles or communities if sentiment signals correlate with demographic or cultural factors; fairness and subgroup analyses are missing.

- Anonymization design is underspecified; potential leakage of identity via writing style or referenced content; no measurement of de-identification effectiveness.

Image Evaluation

Cross-Modal Consistency: 26/50

Textual Logical Soundness: 12/30

Visual Aesthetics & Clarity: 6/20

Overall Score: 44/100

Detailed Evaluation (≤ 500 words):

1. Cross-Modal Consistency

- Major 1: Results “Table” is cited but absent; the core quantitative claim cannot be verified. Evidence: “As shown in Table, the logistic regression model's accuracy...” (Sec. 5)
- Major 2: No figures/tables are provided to depict the proposed pipeline, data flow, or reward workflow; this blocks verification of method–result alignment. Evidence: No figures/tables in manuscript; Sec. 5 references a missing Table.
- Minor 1: Citation/style inconsistencies (e.g., “Vanderbiltget al., 2024”) and typos do not affect meaning.

2. Text Logic

- Major 1: Flagship claim (improving peer-review quality via anonymized feedback and adaptive rewards) is unsupported by experiments that only test Yelp sentiment classification. Evidence: “Experiments... on the Yelp Polarity dataset demonstrate...” (Abstract)
- Major 2: “Bias detection score” is reported (0.0) without a definition, procedure, or dataset attributes; fairness claims cannot be assessed. Evidence: “We also introduce a bias detection score...” (Sec. 4); “bias detection score of 0.0...” (Abstract/Sec. 5)
- Major 3: Reward system efficacy and scalability are asserted without empirical evaluation or ablation. Evidence: “scalability and adaptability... supported by its successful implementation in sentiment analysis tasks.” (Abstract/Sec. 5)
- Minor 1: Notation collision: s is used for “score” and for “submissions $s_1, \dots, s_m$,” risking confusion. Evidence: “each r_i consists of feedback f_i and a score s_i ... submissions $S=\{s_1, \dots, s_m\}$ ” (Sec. 3)
- Minor 2: The mapping from $R_i = \alpha Q_i + \beta C_i$ to operational measures lacks definitions for Q_i and C_i .

3. Figure Quality

- Major 1: Missing results table (Runs 1–3, accuracies) prevents a reader from grasping the intended message; Figure-Alone test fails by absence. Evidence: “As shown in Table...” (Sec. 5)

• Minor 1: No schematic of the anonymized feedback loop or reward pipeline; readers cannot quickly locate components (TF-IDF, sentiment classifier, reward allocator).

Key strengths:

- Clear motivation and timely problem framing.
- Method section provides equations (TF-IDF, logistic regression, reward formula) and reproducible setup (dataset split, TF-IDF=5k, LR, batch, epochs).
- Honest acknowledgment of bias-detection limitation (score 0.0).

Key weaknesses:

- Central claims about peer-review quality and reward efficacy lack direct evidence; experiments are misaligned with the problem domain.
- Missing results table blocks verification of the reported accuracy gains.
- Undefined bias metric and unmeasured reward outcomes.
- Notation inconsistencies and citation/typo noise reduce clarity.

Highest-impact fixes (in order):

- Add the missing results table (runs, metrics, CIs) and a pipeline diagram linking anonymization → sentiment → aggregation → reward.
- Evaluate on peer-review data (real or simulated) with task-relevant outcomes (e.g., meta-review agreement, calibration, bias audits).
- Define and compute the bias metric; include subgroup analyses and debiasing ablations.
- Operationalize Qi and Ci with auditable rubrics; report reward-system experiments (fairness, gaming resistance, scalability).

 Scores

Originality: 2

Quality: 1

Clarity: 2

Significance: 1

Soundness: 1

Presentation: 2

Contribution: 1

Summary

This paper introduces a framework aimed at enhancing the quality of peer reviews in AI conferences by addressing issues of bias and inconsistency. The authors propose a system that integrates anonymized feedback, sentiment analysis, and an adaptive reward mechanism for reviewers. The core idea is to use sentiment analysis on reviewer feedback to mitigate biases and to incentivize high-quality reviews through a reward system that adapts based on the quality of the feedback provided. The authors claim that their approach improves the fairness and efficacy of the review process. The methodology involves converting textual feedback into numerical representations using TF-IDF, followed by sentiment analysis using a logistic regression model. The reward system is designed to allocate points based on the quality of the review, scaled by conference-specific factors. The experimental evaluation primarily focuses on the sentiment analysis component, using the Yelp Polarity dataset to demonstrate improvements in sentiment classification accuracy. The authors report a significant increase in accuracy after applying their anonymized feedback loop. However, the paper also acknowledges a limitation in bias detection, with a score of 0.0 across all experimental runs, indicating that the framework does not explicitly identify and mitigate bias within the sentiment analysis process. While the paper presents an interesting approach to a relevant problem, the experimental evaluation is limited, and the connection between the proposed methods and the specific challenges of peer review is not always clear. The paper's contribution lies in the novel combination of existing techniques to address the problem of peer review quality, but the lack of direct evaluation of the reward system and the limited scope of the experiments raise concerns about the practical applicability of the proposed framework. The paper highlights the potential of using machine learning techniques to improve the peer review process, but further research is needed to validate the effectiveness of the proposed approach in a real-world setting.

Strengths

The paper tackles a significant and timely issue within the AI research community: the need to improve the quality and fairness of peer reviews at conferences. The authors' focus on addressing bias and inconsistency in reviews is commendable, as these issues can undermine the integrity of the academic review process. The proposed framework, which combines anonymized feedback, sentiment analysis, and an adaptive reward system, presents a novel approach to this problem. The use of sentiment analysis to process reviewer feedback is an interesting idea, as it offers a way to quantify and analyze the tone and content of reviews. The authors' claim of improved sentiment classification accuracy after applying their anonymized feedback loop, as demonstrated in their experiments, is a positive finding. The idea of an adaptive reward system that incentivizes high-quality reviews is also promising, as it could encourage reviewers to provide more thorough and constructive feedback. The paper's attempt to integrate these different components into a cohesive framework is a notable strength. The authors also provide a clear description of the methodology, which allows for a good understanding of the proposed approach. The paper's focus on a practical problem and its attempt to offer a technological solution are valuable contributions to the ongoing discussion about how to improve the peer review process. The paper also

highlights the potential of using machine learning techniques to address challenges in academic publishing, which is a relevant and important area of research.

Weaknesses

After a thorough examination of the paper, I've identified several key weaknesses that significantly impact its overall contribution. First, the paper's experimental evaluation is notably weak. The authors rely solely on the Yelp Polarity dataset, a benchmark for general sentiment analysis, to evaluate their approach. This choice is problematic because the dataset is not specific to the peer review context, and it does not capture the nuances and complexities of reviewer feedback. The use of a single, out-of-domain dataset makes it difficult to generalize the findings to real-world peer review scenarios. Furthermore, the paper lacks comparisons with other methods or baselines. The authors only compare against a 'baseline sentiment analysis model without anonymization,' which does not provide a robust assessment of the proposed framework's effectiveness relative to existing or alternative approaches. This absence of comparative analysis makes it challenging to determine the true value of the proposed method. The paper also fails to provide any empirical evidence to support the effectiveness of the adaptive reward system. While the authors describe the reward system in detail, they do not present any experiments that directly measure its impact on reviewer behavior or review quality. This lack of experimental validation raises concerns about the practical utility of the reward system. The paper also acknowledges a significant limitation in bias detection. The bias detection score of 0.0 across all experimental runs indicates that the framework does not explicitly identify and mitigate bias within the sentiment analysis process. This is a critical issue, as the paper's primary motivation is to address bias in peer reviews. The authors themselves note that this highlights a critical area for improvement, as the current framework lacks the capability to explicitly identify and mitigate bias. The paper also suffers from a lack of clarity in the connection between the proposed methods and the specific challenges of peer review. The use of logistic regression for sentiment analysis, while a valid technique, is not particularly novel, and its application to the peer review context is not fully justified. The paper does not adequately explain why logistic regression is the most suitable method for this task, especially given the complexity of peer review feedback. The paper also lacks a clear explanation of how the adaptive reward system is implemented and how it directly addresses the challenges of peer review. The reward system is described in terms of a generic reward allocation formula, but the paper does not provide sufficient details on how the reward parameters are determined or how the system prevents manipulation. The paper also lacks a clear definition of the 'bias detection score' used in the experiments. This lack of clarity makes it difficult to interpret the results and understand the implications of the reported score. The paper's structure and presentation also contribute to its weaknesses. The introduction is overly lengthy and includes excessive citations, which detracts from the main argument. The related work section is organized by technique rather than by contribution to the peer review problem, which makes it difficult to follow the narrative. The paper also includes a large number of marginally relevant citations, which further distracts from the main points. Finally, the paper's claims about the scalability and adaptability of the method are not supported by empirical evidence. The experiments are limited to sentiment analysis on the Yelp dataset and do not demonstrate the method's performance in different conference settings. The paper's claims about the effectiveness of the anonymized feedback loop are also not fully supported by the experiments, as the bias detection score indicates a need for further refinement. In summary, the paper's weaknesses stem from a lack of robust experimental validation, a weak connection between the proposed methods and the specific challenges of peer review, and a lack of clarity in the presentation of the results. These weaknesses significantly impact the paper's overall contribution and raise concerns about the practical applicability of the proposed framework.

Suggestions

Based on the identified weaknesses, I recommend several concrete improvements to strengthen the paper. First and foremost, the authors need to significantly enhance the experimental evaluation. This should include the use of datasets that are specific to the peer review context. The authors could consider creating their own dataset of anonymized reviewer feedback or exploring existing datasets that are more relevant to the task. The experimental evaluation should also include comparisons with other state-of-the-art methods for sentiment analysis and bias detection. This would provide a more robust assessment of the proposed framework's effectiveness. The authors should also conduct experiments to directly evaluate the impact of the adaptive reward system on reviewer behavior and review quality. This could involve A/B testing different reward mechanisms or surveying reviewers about their experiences with the system. The paper needs to provide a more detailed explanation of how the adaptive reward system is implemented and how it prevents manipulation. This should include a clear description of how the reward parameters are determined and how the system ensures fairness and transparency. The authors should also provide a clear definition of the 'bias detection score' and explain how it is calculated. This would improve the interpretability of the results and allow for a more meaningful discussion of the framework's limitations. The paper should also provide a more detailed justification for the use of logistic regression for sentiment analysis in the peer review context. The authors should explain why logistic regression is the most suitable method for this task and discuss its limitations. The paper should also include a more focused and concise introduction that clearly states the problem, the proposed solution, and the main contributions. The related work section should be reorganized to focus on the progression of solutions to the peer review problem, rather than by technique. The authors should also carefully review the citations and remove any that are not directly relevant to the paper's main argument. The paper should also include a more detailed discussion of the limitations of the proposed framework and suggestions for future research. This would demonstrate a more critical and nuanced understanding of the problem and the proposed solution. The authors should also consider conducting user studies or pilot experiments in real-world conference settings to validate the practical applicability of their framework. This would provide valuable insights into the challenges and opportunities of implementing the proposed system in a real-world context. Finally, the authors should revise the paper to address the identified issues with clarity, conciseness, and precision. This would improve the overall readability and impact of the paper. By addressing these weaknesses, the authors can significantly strengthen their paper and make a more meaningful contribution to the field.

Questions

After reviewing the paper, I have several questions that I believe are critical to understanding the proposed framework and its limitations. First, how can the authors justify the use of the Yelp Polarity dataset, which is not specific to the peer review context, to evaluate their approach? What steps were taken to ensure that the results are generalizable to real-world peer review scenarios? Second, what specific methods or techniques were used to detect and mitigate bias in the sentiment analysis process? The paper acknowledges a bias detection score of 0.0, which suggests that the framework does not explicitly identify and mitigate bias. What are the limitations of the current approach, and what are the authors' plans to address this issue in future work? Third, how is the adaptive reward system implemented in practice? What are the specific criteria used to determine the reward parameters, and how does the system prevent manipulation or gaming of the reward

mechanism? Fourth, what is the precise definition of the 'bias detection score' used in the experiments? How is this score calculated, and what does a score of 0.0 indicate? Fifth, why was logistic regression chosen for sentiment analysis, and what are the limitations of this approach in the context of peer review feedback? What alternative methods were considered, and why were they not chosen? Sixth, what are the authors' plans to validate the proposed framework in a real-world conference setting? What steps will be taken to ensure that the system is practical, effective, and ethical? Finally, what are the authors' views on the potential ethical implications of using AI to moderate and incentivize peer review? How can they ensure that the system is fair, transparent, and accountable? These questions are aimed at clarifying the core methodological choices, assumptions, and limitations of the proposed framework. Addressing these questions would significantly improve the paper's overall contribution and impact.

 Scores

Soundness: 1.5

Presentation: 1.75

Contribution: 1.25

Confidence: 3.75

Rating: 2.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights